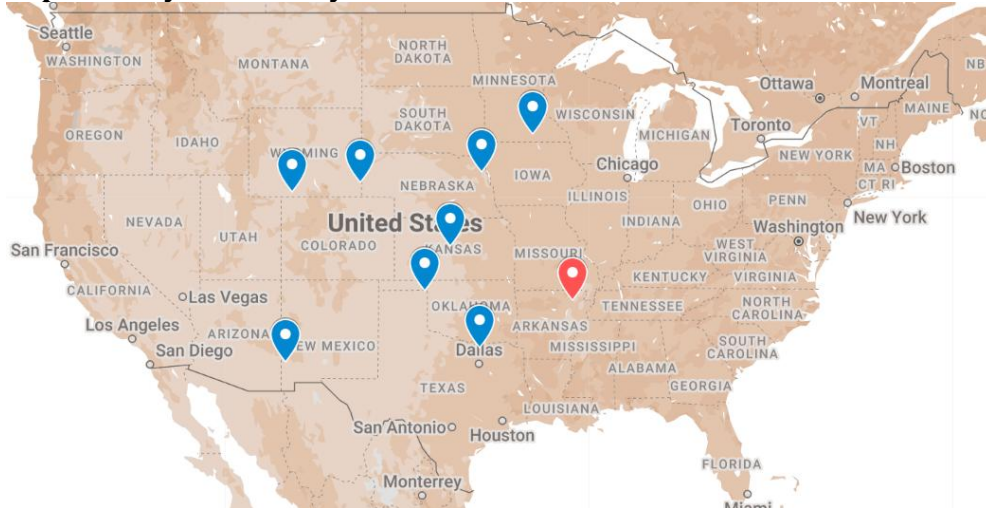


Module 12 – Location Graph

Exploratory Data Analysis



The blue location markers represent stores, and the red marker represents the Current DC.

Model Formulation

MIN:

$$\begin{aligned} & (1-x_1)\sqrt{(41.25-36.04)^2+(-108.49+91.11)^2} + x_1\sqrt{(41.25-38.86)^2+(-109.49+100.99)^2} \\ & + (1-x_2)\sqrt{(42.16-36.04)^2+(-96.8+91.11)^2} + x_1\sqrt{(42.16-38.86)^2+(-96.8+100.99)^2} \\ & + (1-x_3)\sqrt{(43.92-36.04)^2+(-93.62+91.11)^2} + x_1\sqrt{(43.92-38.86)^2+(-93.62+100.99)^2} \\ & + (1-x_4)\sqrt{(38.69-36.04)^2+(-98.69+91.11)^2} + x_1\sqrt{(38.69-38.86)^2+(-98.69+100.99)^2} \\ & + (1-x_5)\sqrt{(36.52-36.04)^2+(-100.31+91.11)^2} + x_1\sqrt{(36.52-38.86)^2+(-100.31+100.99)^2} \\ & + (1-x_6)\sqrt{(41.73-36.04)^2+(-104.27+91.11)^2} + x_1\sqrt{(41.73-38.86)^2+(-104.27+100.99)^2} \\ & + (1-x_7)\sqrt{(33.65-36.04)^2+(-96.83+91.11)^2} + x_1\sqrt{(33.65-38.86)^2+(-96.83+100.99)^2} \\ & + (1-x_8)\sqrt{(32.95-36.04)^2+(-108.93+91.11)^2} + x_1\sqrt{(32.95-38.86)^2+(-108.93+100.99)^2} \end{aligned}$$

Butter Pecan Bluff: (41.24, -108.49)

Fruity Gusher Geyser: (42.16, -96.8)

Ginger Snap Garden: (43.92, -98.6)

Jelly River Delta: (38.69, -98.6)

Pineapple Pop Paradise: (36.52, -100.31)

Praline Park: (41.73, -104.27)

Soda Pop Springs: (33.65, -96.83)

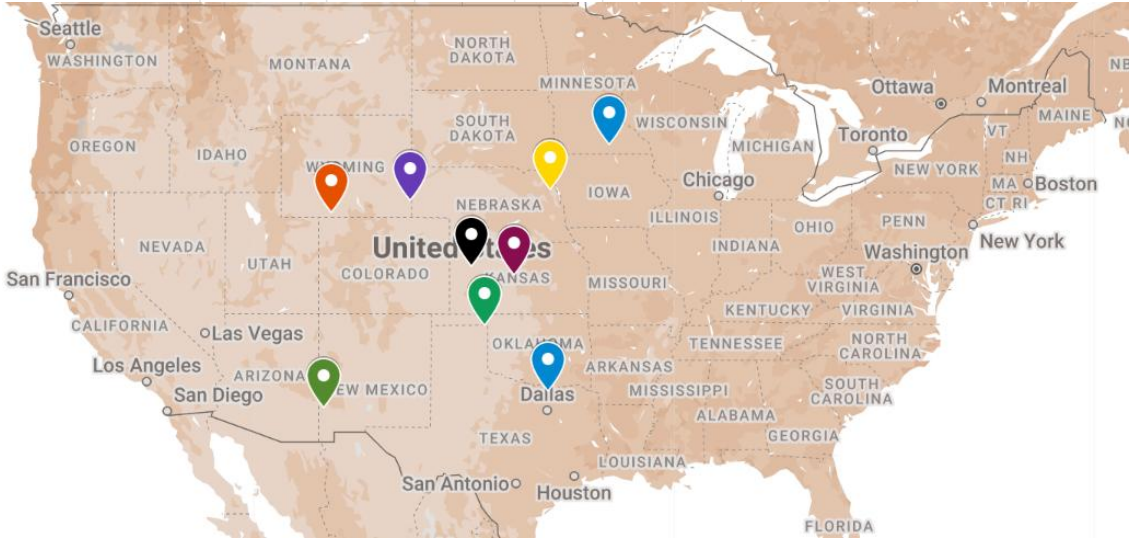
Starburst Starlit Skies: (32.95, -108.93)

Euclidian Distance: $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

Model Optimized for Distance Reduction from DC to Store

This model helps determine whether each retail store should be served by the current distribution center or a proposed new one, to minimize the total travel distance from stores to their assigned distribution center. Using the latitude and longitude of each store and both DCs, it calculates the Euclidean distance and assigns each store to the location that results in the lowest total distance. The model selects the optimal location for the new DC and decides which stores should switch to it, ultimately reducing overall transportation distance and improving logistics efficiency.

							Lat	Long		
		Objective	-591			New DC:	38.86	-100.99		
	Store Location		Current DC		Euclidian	New DC			Model Decision	
Stores	Lat	Long	Lat	Long	Current DC Dist	Lat	Long	New DC Dist	Use New?	Dist
Butter Pecan Bluff	41.25	-108.49	36.04	-91.11	18.14	39	-101	7.87	TRUE	-100.99
Fruity Gusher Geyser	42.16	-96.8	36.04	-91.11	8.36	39	-101	5.34	TRUE	-100.99
Ginger Snap Garden	43.92	-93.62	36.04	-91.11	8.27	39	-101	8.94	FALSE	8.27
Jelly River Delta	38.69	-98.69	36.04	-91.11	8.03	39	-101	2.31	TRUE	-100.99
Pineapple Pop Paradise	36.52	-100.31	36.04	-91.11	9.21	39	-101	2.44	TRUE	-100.99
Praline Park	41.73	-104.27	36.04	-91.11	14.34	39	-101	4.36	TRUE	-100.99
Soda Pop Springs	33.65	-96.83	36.04	-91.11	6.20	39	-101	6.67	FALSE	6.20
Starburst Starlit Skies	32.95	-108.93	36.04	-91.11	18.09	39	-101	9.90	TRUE	-100.99
	38.85875 -100.9925									



This image shows that the colored location markets would use the new DC (black market) and the blue markers would stay with the old DC.

Model with Stipulation

							Lat	Long									
		Objective	76183			New DC:	38.86	-100.99									
	Store Location		Current DC		Euclidian	New DC				Model Decision							
Stores	Lat	Long	Lat	Long	Current DC Dist	Lat	Long	New DC Dist	Use New?	Dist		last_year_demand	expected_yoy_change	new year demand	Weighted Distances		
Butter Pecan Bluff	41.25	-108.49	36.04	-91.11	18.14	38.8588	-100.993	7.8695986	1	-100.9925	1381.98	11%	1534.00	12072			
Fruity Gusher Geyser	42.16	-96.8	36.04	-91.11	8.36	38.8588	-100.993	5.336226	1	-100.9925	1647.83	-8%	1516.00	8090			
Ginger Snap Garden	43.92	-93.62	36.04	-91.11	8.27	38.8588	-100.993	8.9425951	0	8.270097	1742.86	5%	1830.00	15134			
Jelly River Delta	38.69	-98.69	36.04	-91.11	8.03	38.8588	-100.993	2.3086756	1	-100.9925	1850.56	-11%	1647.00	3802			
Pineapple Pop Paradise	36.52	-100.31	36.04	-91.11	9.21	38.8588	-100.993	2.4363	1	-100.9925	1168.47	11%	1297.00	3160			
Praline Park	41.73	-104.27	36.04	-91.11	14.34	38.8588	-100.993	4.3573022	1	-100.9925	1785.98	7%	1911.00	8327			
Soda Pop Springs	33.65	-96.83	36.04	-91.11	6.20	38.8588	-100.993	6.6676445	0	6.199234	1792.63	-5%	1703.00	10557			
Starburst Starlit Skies	32.95	-108.93	36.04	-91.11	18.09	38.8588	-100.993	9.8953137	1	-100.9925	1420.56	7%	1520.00	15041			

My updated model optimizes the location of a new distribution center by minimizing the total demand-weighted distance between stores and their assigned DCs. Instead of

focusing solely on proximity, the model multiplies each store's distance by its projected demand, giving more weight to high-demand locations. This ensures that the distribution strategy prioritizes efficiency not just in distance, but in meeting customer demand, leading to a more cost-effective and operationally balanced solution. In the stipulation, the solver is set to minimize the objective function (weighted distances) by changing variable cells to use or not use the new dc, and I added a constraint for the variable cells to be binary.